

Hyperpartisan Article Classification

Team: G24

1 Introduction

This project focuses on the automatic detection of media bias and polarization in news articles. The main idea behind our work is that how information is presented can influence how readers interpret events and form opinions. Bias is not only expressed through the explicit content of an article, but can also emerge from linguistic choices, such as the words used, the structure of sentences, and the rhetorical strategies adopted by the writer.

We formulated this task as a binary classification problem, where neutral articles belong to class 0, while non-neutral (hyperpartisan) articles belong to class 1.

The choice of this topic was motivated by two complementary interests within the group. On one side, some members were interested in studying information manipulation and understanding how language can contribute to the construction of different news narratives. On the other side, other members wanted to apply statistical machine learning techniques to a real-world Natural Language Processing task.

1.1 Related Work and Research Assumptions

Before developing our methodology, we analyzed existing research in the field of media bias detection. In particular, we focused on two papers that influenced our approach.

The first paper, *A Stylometric Inquiry into Hyperpartisan and Fake News* [1], investigates how stylistic characteristics of texts can be used to identify biased and hyperpartisan content. This work inspired two main assumptions that guided our project.

The first assumption concerns the relationship between political orientation and writing style. The paper suggests that strongly biased articles tend to share common rhetorical patterns, regardless of whether they come from a left or right political perspective. Therefore, we assumed that left- and right-leaning articles are more similar to each other, in terms of writing style, than they are to neutral articles. Based on this assumption, we decided not to distinguish between different political orientations and to merge left and right articles into a single non-neutral category. This allowed us to transform the original problem into a binary classification task: neutral versus non-neutral articles.

The second assumption concerns the structure of news articles. Since the introduction of an article usually contains the main framing of the story and the first

signals of the author’s perspective, we assumed that the beginning of the text contains stronger indicators of bias. For this reason, rather than analyzing the complete articles, we focused only on the first 25% of each article.

The second paper considered in our analysis was *Neural Media Bias Detection Using Distant Supervision with BABE* [2]. This work uses advanced neural models and a large annotated dataset to detect media bias. After studying this approach, we asked ourselves whether comparable results could be achieved using a dataset collected independently through a cheaper and simpler data collection process. This question motivated the construction of our own dataset and the comparison between different machine learning approaches.

1.2 Dataset Construction

We built a custom dataset by scraping news articles from the *AllSides* platform [3], which provides political orientation labels for news sources. Unlike manually annotated datasets, this process is faster and cheaper.

The dataset is structured into triplets, following the BABE dataset structure [2]. Each triplet contains three articles discussing the same macro-topic, with each article representing a different political orientation: left, center, or right. This organization allows us to compare different perspectives on the same topic while reducing the impact of topic-related differences. Consequentially, by construction our classes are imbalanced (2:1). In addition to the training data composed by 978 articles, we also created an external test set composed by 69 articles to evaluate the generalization capabilities of our models on unseen articles.

Although this type of dataset is cheaper to build, it also introduces additional challenges. Since the data comes from real-world news articles collected automatically from the web, it can contain noise, inconsistencies, and variations in writing style. However, this also makes the evaluation more realistic, as real applications must often deal with imperfect data.

1.3 Model Comparison and Objective

Using this dataset, we trained and compared three different models: Logistic Regression, Support Vector Classifier (SVC) and a BERT-based classifier. The main objective of this comparison was to evaluate whether the additional complexity of a transformer-based model provides a significant improvement in performance compared to traditional machine learning approaches.

Overall, the project aims not only to build a media bias detection system, but also to investigate the trade-off between model complexity, performance, and the cost of dataset construction. By working with a self-built dataset, we explore whether effective bias detection is possible even when large manually annotated resources are not available.

1.4 Preprocessing

Besides truncating each article to its first 25% of sentences, as stated before, we removed articles that were too short (fewer than 30 words) and kept only complete triplets of three articles per topic, to avoid unbalanced groups. Next, we applied Named Entity Recognition (NER) to mask person names and nationality/political group mentions, replacing them with a generic placeholder, so that the models would not learn to classify articles based on specific names rather than writing style. From this masked text, we created two versions: one that preserved punctuation and casing (used later for BERT), and one that was cleaned of punctuation and lowercased, which we used to build our stylometric features.

For the stylometric approach, we extracted two types of n-grams. The first type is character n-grams (from 1 to 3 characters), which capture stylistic patterns such as punctuation habits, word endings, and spelling choices, independently of the specific topic of the article. The second type is Part-of-Speech (POS) n-grams (unigrams and bigrams), obtained by tagging each word with its grammatical category (noun, verb, adjective, etc.) and then extracting patterns from these tags. POS n-grams were introduced to try to capture the syntactic structure of sentences rather than their content. Both types of n-grams were converted into numerical features using TF-IDF, with a minimum document frequency threshold to remove very rare patterns.

1.5 Evaluation Methods

All results are reported as macro-averaged F1, which weights both classes equally regardless of support — appropriate given the 2:1 class imbalance:

$$F1_c = \frac{2 \cdot P_c \cdot R_c}{P_c + R_c}, \quad F1_{\text{macro}} = \frac{1}{K} \sum_{c=1}^K F1_c,$$

where P_c and R_c are precision and recall for class c .

To ensure a robust estimate free of data leakage, validation followed a **5-fold nested cross-validation** scheme, with folds pre-defined in the dataset so that articles/topics stay grouped and never appear in both training and test within a fold. Within each outer fold, a `GridSearchCV` with a 4-fold `StratifiedGroupKFold` (grouped by `article_id`) selects hyperparameters maximizing Macro-F1, chosen for its robustness to class imbalance over plain accuracy. Final hyperparameters were determined by *majority vote* across the 5 folds, and the final model was refit on the full training set. Final evaluation was carried out on an **external holdout** set

with non-overlapping topics with respect to training, ensuring a genuine estimate of generalization to unseen topics.

2 Logistic Regression

We firstly trained and validated a **logistic regression** classifier. We compared three text representation strategies: *Bag-of-Words* (BoW), *TF-IDF* and *ELMo embeddings*. Among the three representations, **TF-IDF gave the best and most stable results**, and is therefore the variant reported in details below.

2.1 Theoretical background

Logistic regression models the probability that a document x belongs to the positive class (hyperpartisan article, $y = 1$) through the sigmoid function applied to a linear combination of the features:

$$P(y = 1 | x) = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}}$$

The parameters w and b are estimated by maximum likelihood, which is equivalent to minimizing the **binary cross-entropy** (BCE), also known as log-loss, over the training set:

$$\mathcal{L}(w) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

In our pipeline an $L2$ (Ridge) penalty is added to this objective, with the hyperparameter $C = 1/\lambda$ (inverse regularization strength) selected via grid search:

$$\mathcal{L}_{\text{reg}}(w) = \mathcal{L}(w) + \frac{1}{C} \|w\|_2^2$$

The `liblinear` solver used here optimizes this objective via coordinate descent.

Class imbalance is handled through the `class_weight` parameter, which rescales the contribution of each class in the loss inversely proportional to its frequency ("**balanced**").

A central advantage of logistic regression, and the reason it was chosen alongside SVC and BERT, is **interpretability**: the sign and magnitude of each coefficient w_j indicate the direction and strength of the association between feature j and the predicted probability of hyperpartiticity. In the TF-IDF variant, where every feature corresponds to a token or n-gram, this makes it possible to trace the model's decisions back to concrete linguistic patterns.

2.2 Text representations compared

Bag-of-Words. The simplest representation, encoding each document as a vector of term frequencies (or presences) over the vocabulary, ignoring word order. Used as a baseline to quantify the gain brought by more sophisticated representations.

TF-IDF. Term Frequency–Inverse Document Frequency weighs each term t in document d as

$$\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \log \frac{N}{\text{df}(t)}$$

where $\text{tf}(t, d)$ is the term frequency in the document, N the total number of documents, and $\text{df}(t)$ the number of documents containing t . Compared to plain BoW, TF-IDF down-weights overly common (less discriminative) terms and up-weights rarer terms that recur within specific documents, typically improving linear separability. In the pipeline, the TF-IDF vectorizer was combined via **FeatureUnion** with an explicit stylistometric feature block, and tuned via grid search over n -gram range (unigrams/bigrams), **min_df**/**max_df** vocabulary-filtering thresholds, maximum vocabulary size, and the sublinear term-frequency scaling ($\text{tf} \rightarrow 1 + \log(\text{tf})$).

ELMo embeddings. Unlike TF-IDF and BoW, which yield *sparse*, context-independent representations, ELMo (Embeddings from Language Models) produces *dense*, *contextual* representations via a bidirectional LSTM pre-trained as a language model. In the pipeline, the 1024-dimensional ELMo v3 module from TensorFlow Hub was used, concatenated with the same stylistometric feature block before scaling.

2.3 Stylometric features

Alongside the textual representations, an interpretable set of stylistometric features was computed directly from the raw text and concatenated with the TF-IDF representation before training the classifier.

The feature set includes first-, second-, and third-person pronoun rates; the frequency of exclamation and question marks; the use of parentheses and quotation marks; the presence of ALL-CAPS words; average word and sentence length; and the *type-token ratio*, measuring lexical richness. To better capture ideological framing, we also included lexicon-based features counting the occurrence of intensifiers (e.g., *very*, *extremely*, *totally*), hedges (e.g., *perhaps*, *maybe*, *allegedly*), absolute terms (e.g., *always*, *never*, *everyone*), superlatives, and *loaded language* expressions.

Inspection of the complete TF-IDF logistic regression model revealed that the highest-weight lexical features were largely dominated by outlet-specific boilerplate and byline phrases rather than ideologically meaningful vocabulary. Since news outlet and class label are correlated in the dataset, the classifier is able to exploit these source-identifying artifacts, making the reported performance likely optimistic with respect to generalization to previously unseen publishers. For this reason, the full TF-IDF feature importance is not reported here. Instead, we focus on the stylistometric features, which are largely independent of the news source and therefore provide a more informative and interpretable view of the stylistic signals learned by the model.

Before fitting the logistic regression classifier, all stylistometric features were scaled using

StandardScaler (*with_mean = False*), so each coefficient represents the effect of a one-standard-deviation increase in the corresponding feature. Figure 1 reports the coefficients of the 17 stylistometric features extracted from the final model.

The largest positive coefficients correspond to ideological framing features, particularly *loaded language*, absolute terms, and intensifiers, while general stylistic features have a much smaller impact, suggesting that the classifier relies primarily on rhetorical markers of partisan discourse.

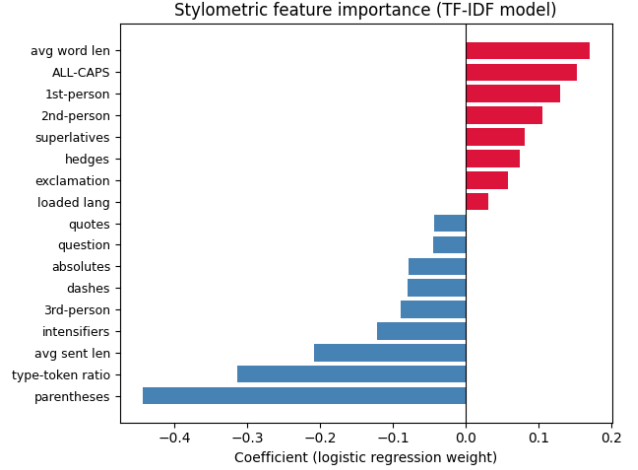


Figure 1: Coefficients of the 17 stylistometric features in the final TF-IDF logistic regression model.

2.4 Results

Model selection and performance estimation were carried out using **nested 5-fold cross-validation**. For each outer fold, an inner grid search evaluated 128 parameter combinations (8 TF-IDF settings $\times$ 16 Logistic Regression configurations). The best configuration selected on the inner folds was then evaluated on the held-out outer fold, yielding unbiased out-of-fold (OOF) predictions while preventing information leakage from hyperparameter tuning.

Fold	Macro F1	Accuracy Best inner-CV	Macro F1
0	0.6789	0.7179	0.6190
1	0.6345	0.6615	0.6071
2	0.6521	0.7020	0.5927
3	0.6365	0.6821	0.6153
4	0.6335	0.6821	0.6239
Mean $\pm$ std: 0.6471 $\pm$ 0.0173		–	–

Table 1: Per-fold Macro-F1 and accuracy for the TF-IDF + Logistic Regression model.

Table 1 reports the Macro-F1 obtained on each outer fold together with the corresponding best inner-CV score. Performance is fairly consistent across folds, with a mean Macro-F1 of **0.6471 $\pm$ 0.0173**, indicating limited variability despite the heterogeneous news topics.

Aggregating the out-of-fold (OOF) predictions across all 5 folds gives an overall Macro-F1 of **0.6476** and accuracy of **0.6892**, with precision/recall of 0.535/0.518 on the minority class and 0.763/0.775 on the majority class. The gap between the two classes’ F1-scores (0.526 vs 0.769) indicates that the model is noticeably more reliable at flagging hyperpartisan content than at correctly ruling it out, a common pattern with class imbalance despite the balanced class weighting.

The final hyperparameters, selected by majority vote across folds, were: $C = 1$ (4/5 folds), `penalty=12` and `solver=liblinear` (5/5 folds), `min_df=3` (5/5 folds), `max_df=0.95` (3/5 folds), `max_features=3000` (3/5 folds), `ngram_range=(1,2)` (3/5 folds), `sublinear_tf=True` (4/5 folds).

The selected configuration was then retrained on the full training set and evaluated on the held-out external test set. Since the test set is relatively small, we complemented the point estimate with a non-parametric bootstrap (2000 resamples) to quantify the uncertainty of the reported Macro-F1 through a 95% confidence interval.

On the holdout test the model obtained a Macro-F1 of **0.6612**, with a bootstrap 95% confidence interval of **[0.5329, 0.7703]** (shown in Figure 2). The CV Macro-F1 falls well within this interval, indicating the consistency of the cross-validation estimate with the holdout performance.

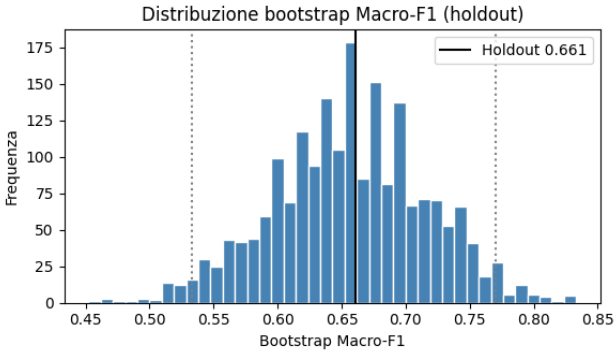


Figure 2: Bootstrap distribution (2000 resamples) of holdout Macro-F1, with observed value and 95% CI bounds.

3 Linear SVC

The initial hypothesis came from *A Systematic Review of Automated Hyperpartisan News Detection* [4], which documents that linear classifiers over shallow lexical and character-level features remain competitive on this task, and that the gap with neural models is often narrower than one would expect.

The Linear SVC occupies an intermediate position, and serves to establish how far a classical, feature-engineered approach can be pushed before a transformer-based model becomes necessary.

3.1 Theoretical background

The Support Vector Classifier (SVC) is a supervised learning algorithm that seeks a decision boundary maximizing the margin between classes. For linearly separable data, the classifier identifies the hyperplane that maximizes the minimum distance from the training samples, leading to improved generalization. When perfect separation is not possible, a regularization parameter C controls the trade-off between maximizing the margin and allowing classification errors. In text classification, documents are typically represented as high-dimensional sparse vectors (e.g., TF-IDF features), making a linear SVC particularly effective. Given a feature vector x , the predicted class is determined by the sign of the decision function

$$f(x) = w^\top x + b,$$

where w is the weight vector and b is the bias term. The model can also be interpreted by inspecting the learned feature weights, providing insight into which textual patterns contribute most to each class.

The classifier optimises a squared hinge loss with L2 regularisation, and we set `class_weight="balanced"` so that the per-sample contributions are reweighted inversely to class frequency.

Our initial pipeline followed the most direct approach available, combining text normalization, semantic embeddings, and hand-crafted stylistic features under the assumption that the latter would provide the primary discriminative signal. Subsequent ablations showed that none of these design choices held up under closer inspection, motivating the revisions described below.

3.2 Text Normalization and Entity Masking

The initial variant applied standard text normalization, lowercasing and stripping punctuation. This configuration consistently underperformed relative to later alternatives, an outcome initially attributed to a pre-processing artifact rather than a substantive finding.

Subsequent analysis indicated that normalization removes precisely the information the model depends on. Capitalization, exclamation marks, parentheses, and quotation marks are not noise with respect to the classification task, but constitute part of the signal itself: a hyperpartisan article and a centrist article covering the same event typically share much of their vocabulary, and the distinction between them is largely encoded in how that vocabulary is punctuated and framed rather than in the vocabulary alone. Removing this surface-level information therefore removes a substantial part of the discriminative signal.

We switched to the masked-only variant, in which only selective NER masking of person and nationality entities are replaced with a placeholder token. We preserved casing, punctuation and sentence structure. Entity masking was retained in this and all subsequent configurations to prevent the classifier from associating specific named actors with a given class, which would

allow it to achieve high performance on the basis of entity co-occurrence rather than genuine stylistic signal. All subsequent configurations were built on this variant.

3.3 Representation Comparison: Word2Vec versus Character n-grams

The next assumption to be tested was that a semantically richer representation would outperform a surface-level one. The comparison instead showed the opposite:

- **Average Word2Vec embeddings** achieved approximately **0.55** Macro-F1. While they capture semantic similarity, averaging compresses each document into a single vector, discarding local stylistic patterns.
- **Character TF-IDF (char_wb)** achieved approximately **0.67** Macro-F1 by preserving morphology, punctuation, capitalization, and function-word usage, all of which are informative stylistic cues for hyperpartisan writing.

This result is consistent with the previous ablation: the discriminative signal lies primarily in writing style rather than semantic content. Consequently, the final pipeline retained **char_wb** TF-IDF features with sublinear term-frequency scaling, while semantic embedding-based representations were discarded.

3.4 Stylometric Features

The final pipeline combines the character n-gram TF-IDF block with the same 17 explicit stylistic features standardised used in the Logistic Regression.

The 17 stylometric features were initially expected to constitute the primary source of discriminative signal. Ablation results did not support this expectation: the stylometric features provided a modest improvement when combined with the character n-gram representation, while a model trained on the stylometric features alone performed substantially worse than the n-gram representation alone. These features therefore complement the n-gram representation rather than driving classification performance independently.

This is consistent with the observation that character n-grams already encode many stylistic cues, such as repeated punctuation and capitalization, making additional hand-crafted features largely redundant.

Part-of-speech n-gram features, evaluated at the same stage, degraded performance and were subsequently excluded from the pipeline.

3.5 Results

Nested cross-validation selected a linear SVC with $C = 0.5$ and **char_wb** character 2–3-grams as the final configuration. The model achieved

Nested CV: F1-macro = 0.6881 ± 0.0116 .

across the outer folds, suggesting stable performance and limited sensitivity to the particular train/validation split. Using this configuration, we then generated out-of-fold predictions over the entire internal dataset for the final performance assessment:

	Precision	Recall	F1	Support
center	0.591	0.595	0.593	326
hyperpartisan	0.797	0.794	0.796	652
macro avg	0.694	0.695	0.694	978
accuracy			0.728	

Table 2: Out-of-fold performance on the internal dataset.

This value of 0.694 is the estimate carried forward, as it falls within the confidence range obtained from the nested cross-validation procedure.

The asymmetry between the two classes deserves further comment. Recall for the hyperpartisan class is 0.79, compared to 0.60 for the center class, indicating that two out of five center articles are misclassified as hyperpartisan. This asymmetry is partly intrinsic to the target definition: hyperpartisan style is a marked phenomenon, characterized by identifiable features such as emphasis, direct address, and absolute statements, whereas the center class is largely defined by the absence of these markers, which is inherently more difficult to detect than their presence. The asymmetry is also partly attributable to data volume, since the center class contains half as many examples as the hyperpartisan class, providing correspondingly less evidence from which the model can characterize it. Balanced class weighting corrects for this imbalance at the level of the loss function, but cannot compensate for the underlying scarcity of information.

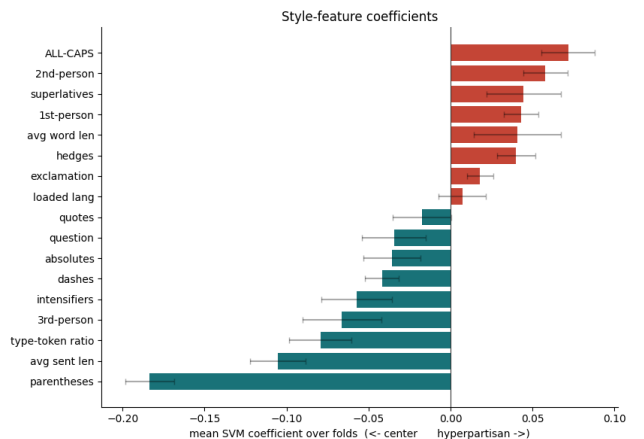


Figure 3: Importance of stylometric features.

3.6 Feature Importance

Aggregating the weight vector across the outer folds gives a stable and readable picture. As shown in Figure 3, pushing toward hyperpartisan: capitalisation,

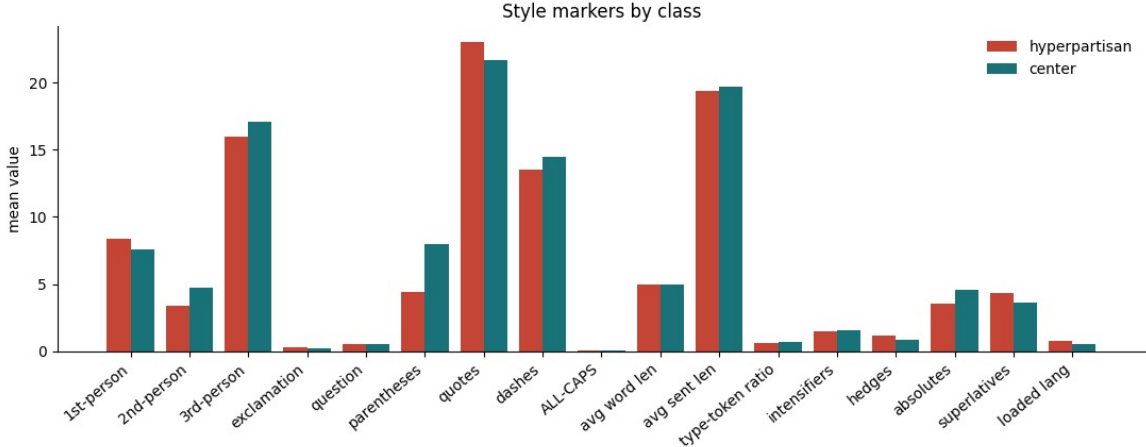


Figure 4: Mean value of the 17 stylistic features by class.

2nd- and 1st-person pronouns, superlatives, hedges. Pushing toward center: parentheses by a wide margin, then average sentence length, type-token ratio, 3rd-person pronouns. The narrow error bars indicate that the coefficient estimates are stable across folds. It is a coherent portrait: partisan writing addresses the reader directly and speaks in emphatic absolutes, while neutral reporting is longer-sentenced, lexically varied, third-person, and parenthetical.

And then the second Figure 4 undercuts the first, which is the most instructive thing we found. Considered individually, the stylometric markers show limited separation between the two classes. Class means for capitalization, exclamation mark frequency, and first-person pronoun rate are nearly identical between classes; only parentheses and type-token ratio exhibit a visible difference. The differences the model relies on are, at the level of individual features, minuscule.

Taken together, these two results indicate that the stylometric features carry discriminative value primarily in combination rather than in isolation. A large coefficient does not correspond to a large difference between class means; rather, it indicates that, conditional on the remaining feature vector, that dimension contributes to shifting the decision boundary. The model therefore exploits a joint pattern distributed across many individually weak signals, rather than relying on a small number of strongly discriminative features. This also accounts for the limited performance of the stylometric features when used in isolation: no single stylistic marker is individually decisive, and the corresponding signal, while genuine, is diffuse across the feature set.

We deliberately did not plot the character n-gram coefficients. With tens of thousands of two- and three-character features, individual weights are not interpretable and would support no meaningful claim.

3.7 Discussion and Limitations

While requiring only seconds of training time on CPU and remaining fully interpretable, the limitations of this approach are apparent. The model has no access to word order, syntax, or context beyond a three-character

window; it cannot distinguish a partisan phrase that is quoted from one that is asserted; and it performs comparatively poorly on the center class. These limitations correspond directly to capabilities a contextual model would be expected to provide, which motivates using this configuration as the reference point for the cost-benefit comparison central to this project.

4 BERT-based model

Taking inspiration again from *A Systematic Review of Automated Hyperpartisan News Detection* [4] we decided to implement a BERT-based model. Specifically we fine-tuned **bert-base-cased** [5] and tested different architectural choices. We explored three axes of variation: the sequence pooling strategy, the input representation (adding features) and the training regime (full fine-tuning versus partial layer freezing). We also report an overfitting analysis that motivated our checkpoint-selection procedure.

4.1 Final architecture

First we show the final architecture that we decided upon after the various experiments. The pooled representation z is passed through a classification head. We compare a minimal linear head,

$$\hat{y} = Wz + b, \quad W \in \mathbb{R}^{2 \times d'},$$

against a small MLP head with a non-linearity and normalization in between,

$$\hat{y} = W_2 \text{Dropout}_p(\text{LayerNorm}(\text{GELU}(W_1 z + b_1))) + b_2,$$

where d' is the pooled representation’s dimensionality. Given the class imbalance between biased and neutral articles, we minimize a class-weighted cross-entropy loss:

$$\mathcal{L} = - \sum_{c \in \{0,1\}} w_c y_c \log \text{softmax}(\hat{y})_c,$$

with balanced weights

$$w_c = \left(\frac{N}{K \cdot n_c} \right)^\alpha,$$

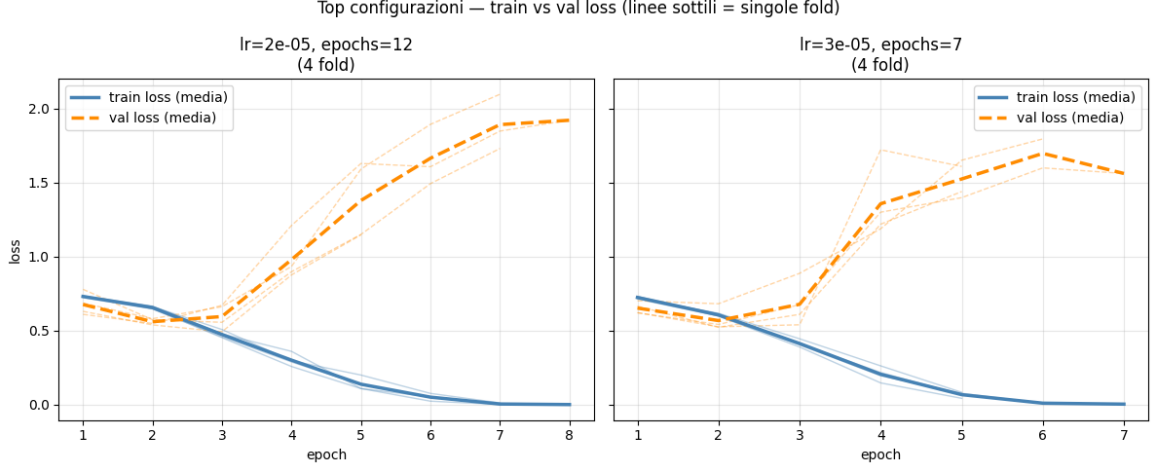


Figure 5: Training and validation loss per epoch of the 3 best hyperparameter choices. Validation loss reaches a minimum around epoch 2–3 before increasing, motivating best-checkpoint selection over taking the final epoch.

where N is the total number of training examples, $K = 2$ the number of classes, n_c the number of examples of class c , and $\alpha \in (0, 1]$ a damping exponent ($\alpha = 1$ recovers the standard inverse-frequency weighting used in our baseline configuration).

4.2 Sequence Pooling Strategies

Let $H = (h_0, h_1, \dots, h_n) \in \mathbb{R}^{(n+1) \times d}$ denote the final hidden states produced by BERT for a tokenized input of length n (plus the [CLS] token h_0), and let $m \in \{0, 1\}^{n+1}$ be the attention mask. We compare three ways of collapsing H into a single document-level representation z :

CLS pooling. The standard approach uses the final hidden state of the [CLS] token:

$$z_{\text{CLS}} = h_0.$$

Mean pooling. We instead average the hidden states of all non-padding tokens:

$$z_{\text{mean}} = \frac{\sum_{i=0}^n m_i h_i}{\sum_{i=0}^n m_i}.$$

This treats every token’s contribution equally, which is preferable when the discriminative signal is distributed across the sequence rather than concentrated in a single position.

Concatenated pooling. We also tested the combined representation:

$$z_{\text{cat}} = [z_{\text{CLS}}; z_{\text{mean}}] \in \mathbb{R}^{2d}.$$

This very last pooling is the one that we chose for the final version.

4.3 Overfitting Analysis and Checkpoint Selection

Plotting training and validation loss per epoch reveals the canonical overfitting signature: training loss decreases monotonically, while validation loss reaches a minimum around epoch 3–4 and increases thereafter (Figure 5). We therefore select, for each fold, the checkpoint at the epoch of minimum validation loss (or close to it) rather than the final epoch:

$$e^* = \arg \min_{e \in \{1, \dots, E\}} \mathcal{L}_{\text{val}}(e).$$

4.4 Input representation

Inspired by prior work ([4] team 122) combining contextual embeddings with explicit linguistic features [6], we hypothesized that concatenating BERT’s pooled representation with a hand-crafted feature vector $f \in \mathbb{R}^{58}$ of 8 stylistic features (punctuation density, negation and pronoun frequency, modal verb usage, lexical diversity) and 50 part-of-speech bigram TF-IDF features, would inject complementary info that could benefit the classification.

First attempt We first extracted frozen embeddings from an already fine-tuned mean-pooling model, concatenated them with f , and trained a separate shallow classifier (LinearSVC) on top:

$$\hat{y} = \text{SVC}([z_{\text{mean}}; f]).$$

Second attempt We restructured the model so that f is concatenated *inside* the forward pass and the entire network — backbone, features, and head — is trained jointly:

$$\hat{y} = \text{MLP}([z_{\text{mean}}; f])$$

This required a custom data collator to batch token sequences and fixed-size feature vectors together.

4.5 Layer Freezing Experiment

Transfer learning theory suggests that lower Transformer layers encode general-purpose linguistic knowledge (morphology, syntax), while higher layers specialize toward the pre-training objective and, after fine-tuning, toward the downstream task [7]. Freezing the lower k of the $L = 12$ encoder layers,

$$\theta_\ell \text{ frozen for } \ell \leq k, \quad \nabla_{\theta_\ell} \mathcal{L} = 0 \text{ for } \ell \leq k,$$

is a standard mitigation against catastrophic forgetting when fine-tuning on small corpora. We tested $k = 8$ against full fine-tuning ($k = 0$).

4.6 Results

The results in terms of $F1_{\text{macro}}$ score on the validation set show an improvement due to the pooling strategy: the CLS token-based one achieves the lowest performance, with an $F1_{\text{macro}}$ of 0.694. Mean pooling bringing the $F1_{\text{macro}}$ to 0.737. Finally, the concatenation strategy proves to be the most effective, reaching a $F1_{\text{macro}}$ of 0.757, making it the optimal choice for the final model among those tested for this classification task.

The variations on the model that add a feature vector f only decreased the performance: $F1_{\text{macro}} = 0.687$.

Contrary to the theoretical motivation, freezing less layers consistently *hurt* performance: $F1_{\text{macro}}$ dropped from 0.757 to 0.608 for mean pooling, and similarly for the concatenated variant.

Table 3 summarizes whole the progression across pooling strategies, the layer-freezing ablation, and the final tuned configurations.

Table 3: $F1_{\text{macro}}$ across BERT configurations. CV = 5-fold cross-validation (out-of-fold)

Method	$F1_{\text{macro}}$ (CV)
Pooling z_{CLS}	0.694
Pooling z_{mean}	0.737
Pooling z_{cat}	0.757
With feature vector f	0.687
Freezing first 8 layers	0.603
Freezing all 12 layers	0.757

Conclusions

4.7 Results on the external test set

We evaluated the various models on the external test set.

Logistic Regression The final model obtains a Macro-F1 of **0.6612** and accuracy of **0.6957** (Figure 6), with a bootstrap 95% confidence interval of **[0.5329, 0.7703]** Figure 2. The CV Macro-F1 (0.6476) falls well within this interval, indicating that the cross-validation estimate is consistent with the holdout performance.

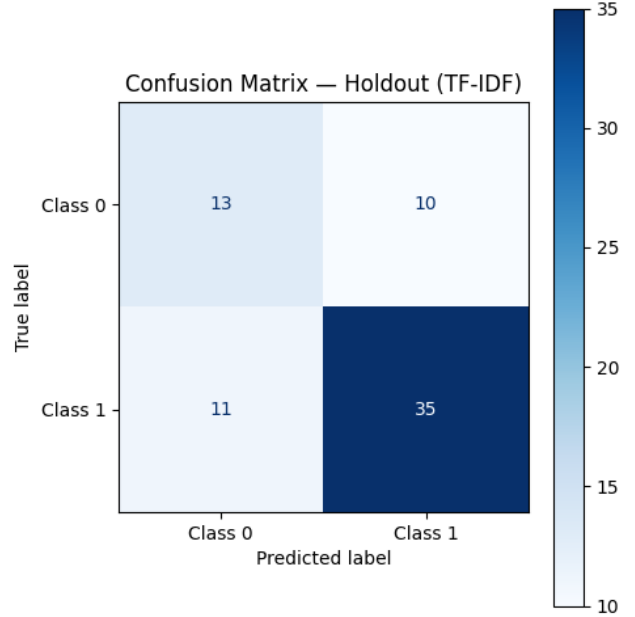


Figure 6: Confusion matrix on the external holdout set on the Logistic Regression.

SVC Performance on the external test set attested itself at $F1_{\text{macro}} = 0.713$. This is close to the internal cross-validation estimate (0.694), and the error structure was consistent between the two evaluations: recall for the hyperpartisan class was 0.78 on the external set compared to 0.79 internally, and recall for the center class was 0.65 compared to 0.60.

Table 4: Performance on the external held-out set.

Class	Precision	Recall	F1
Center	0.600	0.652	0.625
Hyperpar.	0.818	0.783	0.800
Macro Avg	0.709	0.717	0.713

BERT The concatenated pooling with an MLP head achieved the best result overall. Its external test score $F1_{\text{macro}} = 0.872$. This result seemed too far away from the CV one even after putting in place checkpoint to avoid overfitting. So in this specific single case we evaluated the model once again on an external dataset but we made it larger. The expanded test set to a larger has $n = 246$ articles (82 topics). The estimate became more conservative: $F1_{\text{macro}}$ dropped from cross-validation’s 0.757 to 0.715. We report this revision transparently: it illustrates that a small held-out set can itself be a source of *optimistic* bias due to sampling variance.

4.8 Are the models better than just guessing?

What would a non-learning classifier score on this task? Given the class imbalance $\frac{2}{3}$ hyperpartisan articles and $\frac{1}{3}$ neutral ones, let $\epsilon \rightarrow 0$ be the probability of ever predicting the minority class. Assuming predictions

and labels’ probabilities to be independent this is the table of the joint distribution

	label = 1	label = 0
pred = 1	0.33ϵ	$0.33(1 - \epsilon)$
pred = 0	0.67ϵ	$0.67(1 - \epsilon)$

Then the macro-F1 is: $\frac{1}{2} \left(2 \frac{0.33\epsilon}{0.33+\epsilon} + 2 \frac{0.67(1-\epsilon)}{0.67+0.67(1-\epsilon)} \right)$
 With $\epsilon \in (0, 1)$ this looks like Figure 7:

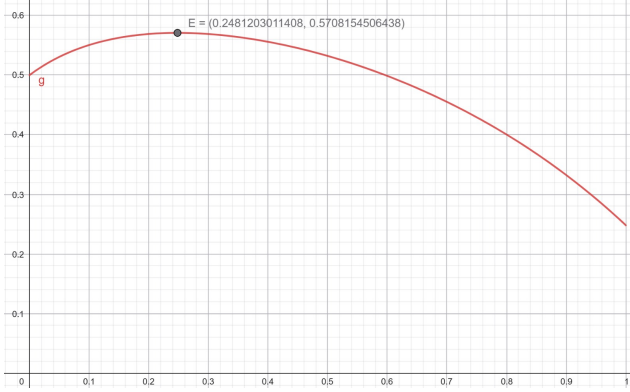


Figure 7: Form of the Macro-F1 function

This means that as long as our models do better than 0.57 of Macro F1 then randomly guessing is always worse, regardless of the specific probability for the prediction of the minoritarian class ϵ . In the case of guessing 1 following a Bern($\frac{2}{3}$) distribution the specific Macro F1 threshold is 0.5.

All our models clear these thresholds on the test set, some with more margin than others.

Model Comparisons

Model	Val F1 _{macro}	Test F1 _{macro}
Log. Reg.	0.6476	0.6612
SVC	0.694	0.713
Bert	0.757	0.715

Table 5: Final models comparison

We can contextualize all the results thanks to the review *A Systematic Review of Automated Hyperpartisan News Detection* (Maggini, Bassi, Piot, Dias & Gamallo Otero, 2025). The logistic regression and the SVC, with respect to models that likewise only leverage semantical features, rank in the medium-low end of the spectrum. Instead Bert behaves better even in the rankings although we need to point out the evaluations on this review stem from different datasets so there isn’t the same assurance as using the same benchmark. Still we observed that deep learning models outperform traditional machine learning approaches on this task even if not by a big margin. Of course, remembering that our final result for BERT is done on a larger (and so more diverse) dataset.

Masking effects As stated in the preprocessing named entities of type PERSON and NORP (nationalities, religious or political groups) are replaced with a generic placeholder token ENT. Empirically, BERT’s F1 is comparable with and without masking, while the SVC baseline’s performance changes substantially under the same ablation — suggesting that BERT’s contextual representations are less dependent on this particular form of surface regularization.

Feature analysis Both the Logistic Regression and the SVC models agree strongly on the overall ranking and direction of stylistic features driving classification. Parentheses, long average sentence length and the use of the 3rd person pronouns are the strongest predictors pulling toward the center class, while ALL-CAPS, 1st and 2nd person pronouns and superlatives are among the strongest predictors pulling toward the hyperpartisan class. This intuitively checks out with the intrinsic nature of the task: hyperpartisan style is a phenomenon characterized by emphasis, direct address, and absolute statements. The near-zero coefficients also line up: loaded language, exclamation marks, and quotes sit close to the boundary in both models. The main differences is in magnitude. This interpretability is lost once we deploy BERT but since it does score better then the previous models we can deduce that the added semantic context weights in. Our failed fusion attempt suggests that a fully fine tuned BERT already internalizes much of the stylistic signal that hand-crafted linguistic features were meant to make explicit.

This gap also points to a limitation these models share. Research on hyperpartisan detection distinguishes lexical bias, stemming from word choice and phrasing, from informational bias, stemming from which facts are selected, omitted, or emphasized relative to an article’s broader context (Fan et al., 2019).

Final reflections

Finally, our evaluation rests on a small test set, which limits how much confidence we can place in the reported performance differences. With so few examples, individual misclassifications carry outsized weight on the final metrics, and the gap between models — including the modest edge BERT shows over the traditional classifiers — could shrink or reverse on a larger sample. A more robust comparison would require scaling up the test set, ideally alongside cross-validation or bootstrapped confidence intervals, before drawing firm conclusions about which approach generalizes best.

Our experiments with BERT validated the limited perspective provided by the F1 score as argued by the paper [2]. Specifically (Figure 8): while the loss decreases through the first few epochs and then rises the F1-macro metric does not mirror this behaviour. It continues to trend upward reaching its plateau (≈ 0.73 – 0.74) at exactly the epochs where the loss is at its worst (epoch 6–8), rather than degrading alongside it, not a standard overfitting case. Since the F1 score is chosen as the primary evaluation criterion across different papers

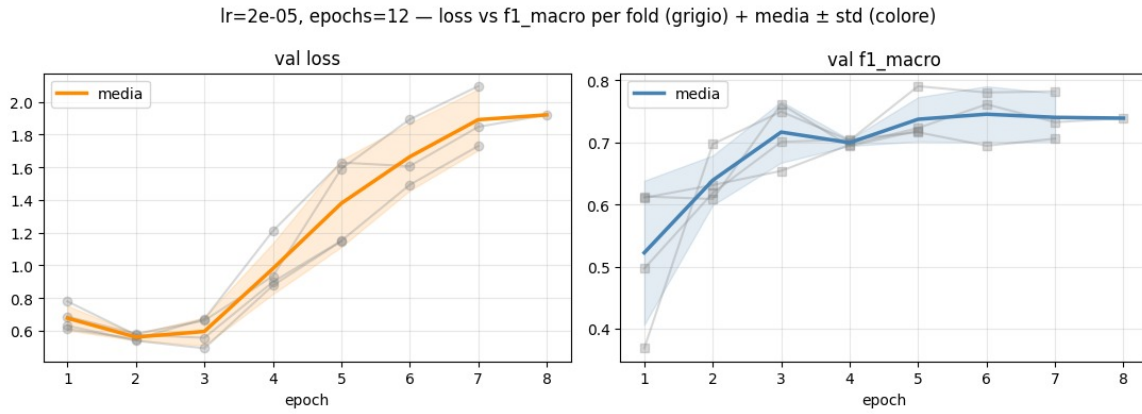


Figure 8

and overviews this observation highlights the need for other measures. As discussed in (Spinde et al., 2021) there’s a need for a more granular evaluation scheme, decomposing the task into sub-tasks along the lines of CheckList (Ribeiro et al., 2020), to assess performance separately on lexical, informational, and contextual bias rather than relying on one overall score.

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